Ali **Marjaninejad**

Applied Scientist · AIML | GenAI | Physical AI

U.S. Permanent Resident | Seattle, WA, USA

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AI/ML applied scientist with a published track record in GenAI, Neuroscience, and Physical AI

Professional Experience

amazon — *Seattle, WA*

Applied Scientist – Gen AI — *Jan. 2025 - Present*

- Designed and published reasoning-based **multimodal benchmarking datasets** and evaluation metrics for next-generation MM-LLMs (**CVPR 2026 Highlight paper**).
- Developed and **productionized** end-to-end **LLM-based automation** of multimodal manual workflows—incorporating agentic capabilities, MMRAG, confidence estimation, and performance evaluation—driving **multi-$M annual OpEx savings**.
- Architected **multi-agent synthetic data generation** for low-resource classification at ∼**50**× **cost reduction**.
- Developed plug-and-play **active learning** wrapper for selective prediction, cutting annotation costs ∼**30%** on 100+ tasks.
- Built user-facing self-serve **MLOps tooling**, minimizing human-in-the-loop overhead for ML-based automation pipelines.

ceribell® — *Sunnyvale, CA*

Staff Data Scientist — *Feb. 2022 - Jan. 2025*

- Led **cross-functional** projects across clinical, hardware, and commercial teams from ideation through **FDA submission**, resulting in **FDA approval**.
- Applied **AI** and classical **Signal Processing** techniques to process, enhance, and make inferences from **biological signals**, producing meaningful insights into pipeline performance and generating actionable statistics.

USC University of Southern California (USC) — *Los Angeles, CA*

Postdoctoral Research Fellow — *Jul. 2021 - Feb. 2022*

- Developed **life-long learning**, autonomous **bio-inspired robotic systems** with embodied intelligence, resulting in publications in **Nature Machine Intelligence**, **Science Advances**, and **IEEE/RSJ IROS**.
- Utilized AI and machine learning techniques to model and make inferences from **bio-signals** (Brain Computer Interface devices, etc.).

Graduate Education

USC University of Southern California (USC) — *Los Angeles, CA*

- Ph.D. Biomedical Engineering, GPA: 3.95/4 | Research focus: AI in Bio-Signal processing and Bio-robotics — *2021*
- M.Sc. Electrical Engineering, GPA: 3.88/4 | Track: Data Science — *2017*

Amirkabir University of Technology (AUT) — *Tehran, Iran*

- M.Sc. Biomedical Engineering, GPA: 4/4 | Track: Signal Processing — *2015*

Select Publications

20+ peer-reviewed publications across **top-tier journals and conferences**, including:

1. "RMIR: A Benchmark Dataset for Reasoning-Intensive Multimodal Image Retrieval." **CVPR (Highlight Paper)**, 2026. (link)
2. "Autonomous functional movements in a tendon-driven limb via limited experience." **Nature Machine Intelligence**, 2019. (link)
3. "Biological underpinnings for lifelong learning machines." **Nature Machine Intelligence**, 2022. (link)
4. "Curriculum is more influential than haptic feedback when learning object manipulation." **Science Advances**, 2025. (link)
5. "A bio-inspired framework for joint angle estimation from non-collocated sensors in tendon-driven systems." **IEEE/RSJ IROS**, 2020. (link)

Technical Skills

Gen AI	Multimodal LLMs, RAG, Embedding, Distillation, Agentic AI, MCP, Responsible AI, Orchestration, Evaluation/Benchmarking, Confidence Estimation, AWS Bedrock
ML	Deep Learning, Reinforcement Learning, Active Learning, Transfer Learning, Supervised and Unsupervised Learning, Machine Vision, Classification, Optimization, Statistics, Time Series Analysis, Dimensionality Reduction, Exploratory Data Analysis, Feature Engineering
Domain	AWS, Large-Scale ML Pipelines, MLOps, Big Data, Benchmarking, Robotics, Physical AI, Physics Simulation (MuJoCo), Lifelong-Learning Machines, Bio-Signal Processing, Biomedical Hardware Design
Programming	Python, C/C++, Assembly, HTML, MATLAB, R
Languages	English, Persian, Turkish